Lab 6 — Branch Coverage Report

Software Engineering Lab | shipping_quote.py | SETAPESU26/L6_L7_04_shipping_quotation

Objective

Measure and improve branch coverage of shipping_quote.py — confirming that every if / elif / else decision point evaluates to both True and False across the test suite, not merely that each line executes once.

Tools and Commands

- coverage.py 7.16.0 run with the --branch flag, pytest 9.1.1
- Run: python -m coverage run --branch -m pytest test_shipping_quote.py
- Report: python -m coverage report -m

Baseline Branch Coverage

The starter test alone exercised only the True side of most conditions. coverage.py reported 36 branch exits in shipping_quote.py, 10 of which were only partially covered (BrPart = 10), meaning one direction of each decision was never taken.

Module	Branches	Partial	Coverage
shipping_quote.py	36	10	45%

Decision Points Targeted

Building on the line-coverage test suite (15 tests), two further test cases were added to close the remaining branch gaps and make the intent explicit:

- Hazmat + non-fragile combined with weight_kg > 30, to confirm the surcharge and heavy-weight branches compose correctly
- A pressure log where a low reading is followed by a stable reading, confirming depressurization_streak resets to 0 (the else branch) rather than only ever accumulating

Together with the line-coverage suite, both directions of every decision are now exercised:

- isinstance(shipments, list) and handling_fee < 0 — both pass and fail
- manifest_approved — both True and False
- transport_mode — air, ground, and an unrecognised mode
- weight <= 0 — both the rejected and accepted paths
- is_hazmat combined with is_fragile — all four combinations reachable
- weight > 30 — both the heavy-weight surcharge and the standard rate
- express_priority — both True and False
- rejected_parcels > 3 — both FLAGGED_REVIEW and the normal path
- total_quote == 0.0 — both NO_CARGO and CLEARED/FLAGGED_REVIEW
- psi < 14.0 and depressurization_streak >= 2 — leak detected, streak reset, and stable cases

Final Result

With seventeen total test cases, every branch in shipping_quote.py is fully covered in both directions.

Module	Branches	Partial	Coverage
shipping_quote.py	36	0	100%
test_shipping_quote.py	0	0	100%
TOTAL	36	0	100%

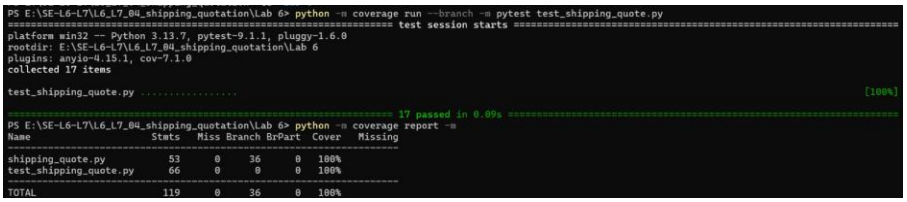


Figure 1. Terminal output — 17 tests passing, 100% branch coverage (BrPart = 0).

Conclusion

Branch coverage improved from 45% (10 partially-covered branches) to 100% (0 partial), exceeding the 90% target. Every decision point in shipping_quote.py is now confirmed to behave correctly along both of its possible paths.